

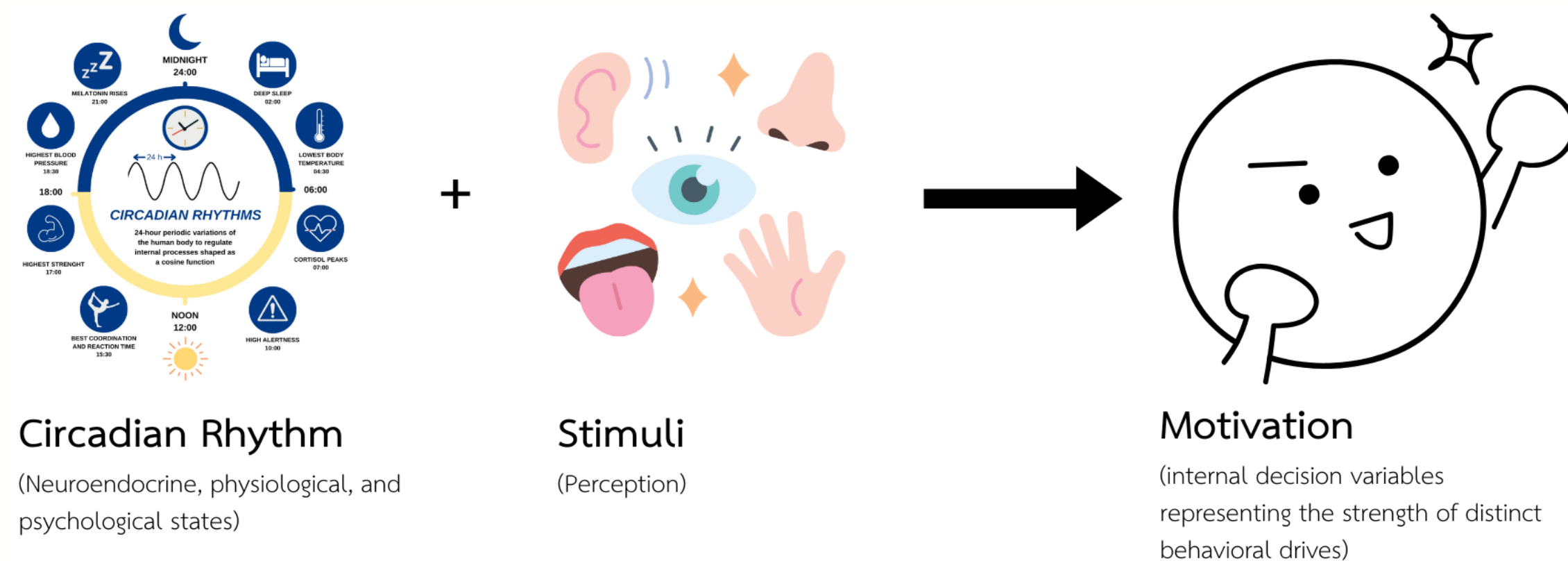


Simulation of “*Maternal Instinct*”

Abstraction

Maternal care is a **foundational** form of **altruism**, yet the mechanisms by which prosocial strategies emerge and persist remain unclear. This thesis proposes a computational framework that **integrates physiological, psychological, and neuroendocrine-inspired internal states dynamics within an evolutionary agent-based model** to investigate the emergence of “**maternal altruism**”. Mother agents operate through a hierarchical cascade from genetic traits -> perception -> internal state modulation -> motivation -> action, in which **maternal bond strength regulates** stimulus interpretation through **allostatic** control. Child agents are immobile and communicate exclusively via distance-dependent distress signals, creating an asymmetric caregiving dynamic. Using **spatial grid-world simulations** with **multiple family units**, mothers inherit **genetically encoded traits** and populations evolve under varying levels of resource availability, predation pressure, and population density. This **research explores how maternal caregiving strategies emerge** through evolution rather than explicit programming, advancing understanding of **cooperation, prosocial AI design, and AI safety**.

Motivation Selection in an Organism



Current Progress

Implemented a mother & child agent architecture with coupled physiological, psychological, and neuroendocrine states, and **validated adaptive motivation switching** (Protect child, Care, Forage, Self-preservation) under varying threat and resource conditions.

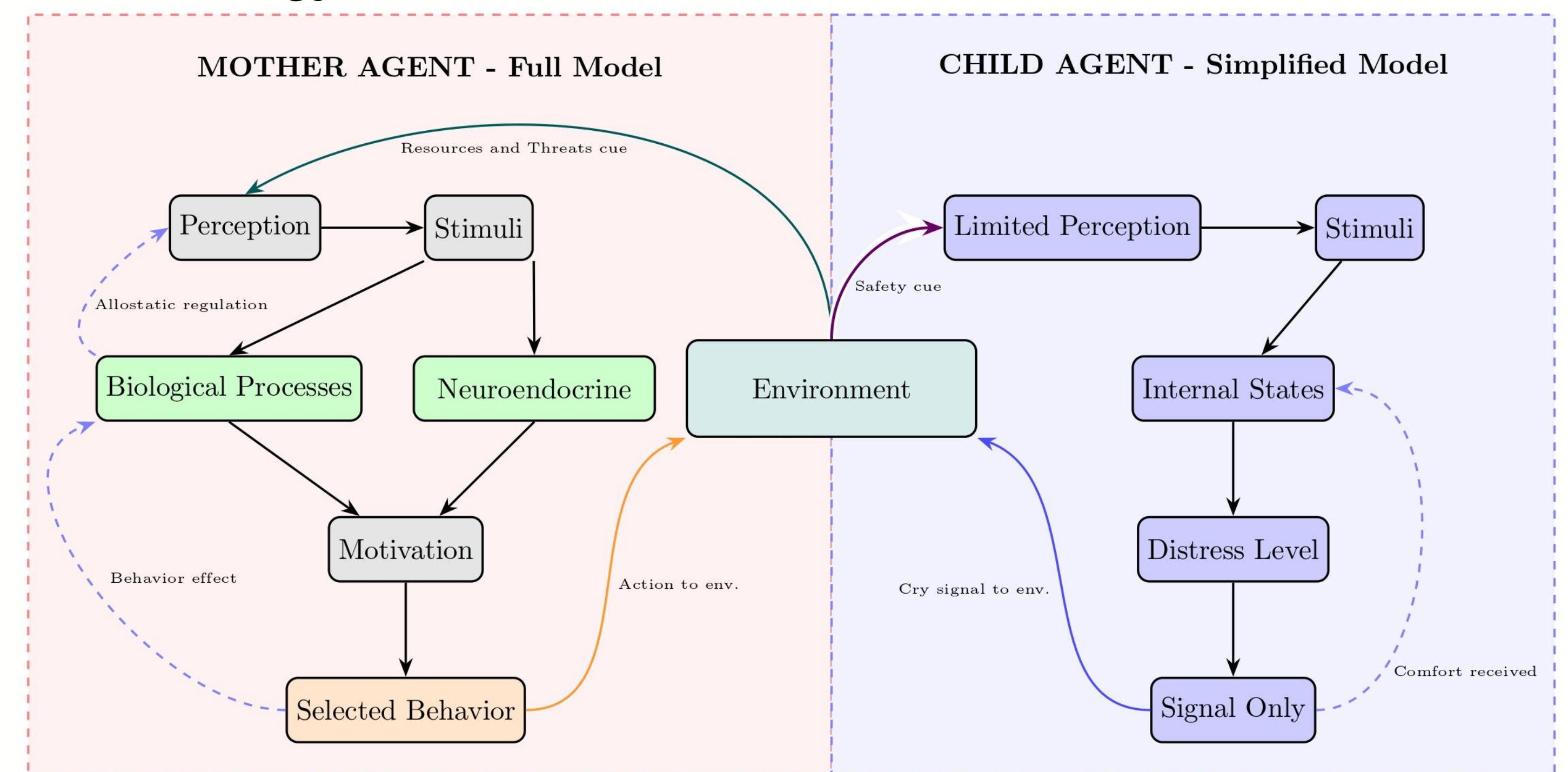
Next Steps

Extend simulations across generations to quantitatively **analyze the evolutionary stability of maternal caregiving** strategies under varying environmental pressures.

Objectives

To **develop** and **analyze** an **evolutionary agent-based model** that integrates physiological, psychological, and neuroendocrine-inspired states to explain **how maternal altruistic caregiving emerges** and persists under environmental pressures.

Methodology



Scenario Test

